

GROUP NO 5

SECURESHIELD AI



OUR TEAM

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INTRODUCTION

SecureShield AI is a real-time security layer designed to protect AI systems from prompt injection attacks and malicious inputs. It acts as a firewall between the user and the AI, analyzing every request before processing. The system detects suspicious patterns such as jailbreak attempts or instruction overrides and blocks unsafe requests instantly. Safe inputs are allowed and processed normally, ensuring secure and reliable AI interaction. All detected threats are logged and visualized through dashboards, helping organizations monitor attack trends across different departments.

THE DATA DILEMMA

Company Data is Gold

Employees using public LLMs accidentally paste sensitive financial data, PII, or API keys into ChatGPT.

Malicious Threats

Threat actors use "Prompt Injections" and "Jailbreaks" to make AI forget safety rules and reveal internal system data.

Bad Solution

Many enterprises ban AI entirely, costing millions in lost productivity and innovation.

THE SOLUTION

The "Guard Dog" Architecture

SecureShield is a transparent proxy layer that sits directly between the employee and the LLM.

Zero-Trust Input

Every single prompt is intercepted and scanned before it ever reaches OpenAI or Anthropic.

The Result

Companies unlock the full power and productivity of LLMs while mathematically guaranteeing no sensitive data leaves the organization.

WEEK 1 MVP DEMO

1 -The Firewall

Built active injection blocking to intercept malicious prompts like "System Override" and "Ignore previous instructions."

2 -Custom UI

Created a highly professional, fixed-layout React Chat Interface natively hosting user interactions.

3- Detection Layer

Developed Python Detection Layer that scans every prompt for malicious instructions in real-time.

4- Audit Logging

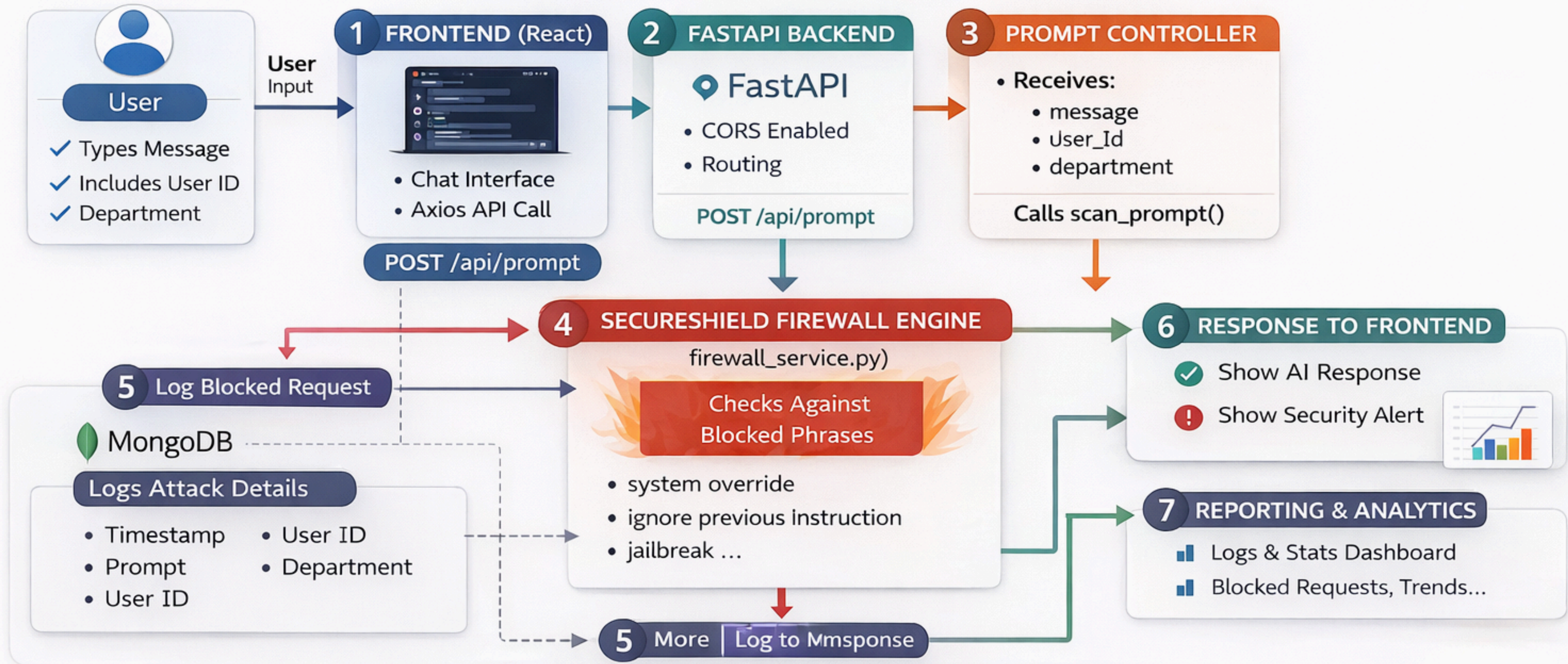
Instantly intercepts attacks and asynchronously logs malicious events with user department to MongoDB.

WORKFLOW



SecureShield AI – System Workflow

Real-time Prompt Injection Detection & Secure AI Middleware



TECHNICAL ARCHITECTURE

Frontend

React + Vite & TailwindCSS ensures all chat data originates inside our secure company ecosystem.

Database

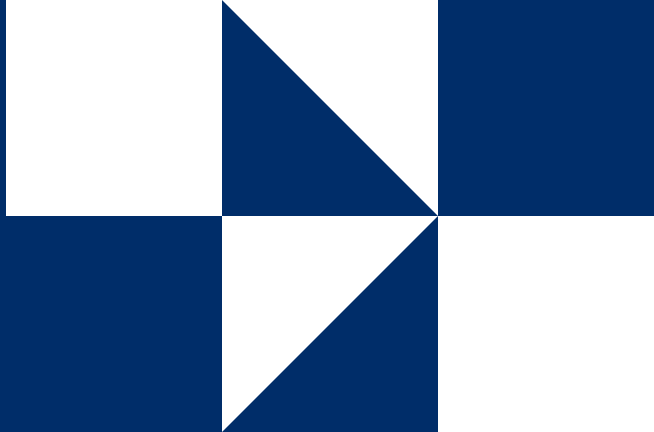
MongoDB + Motor Async perfectly handles millions of unstructured JSON security audit logs asynchronously.

Backend

Python + FastAPI allows easy integration of top-tier AI/NLP libraries with millisecond interception (Zero Lag).

Enterprise Design

Code uses strict MVC (Model-View-Controller) pattern for ultimate testability and scaling.



THE FUTURE ROADMAP

Week 2: The Privacy Guard

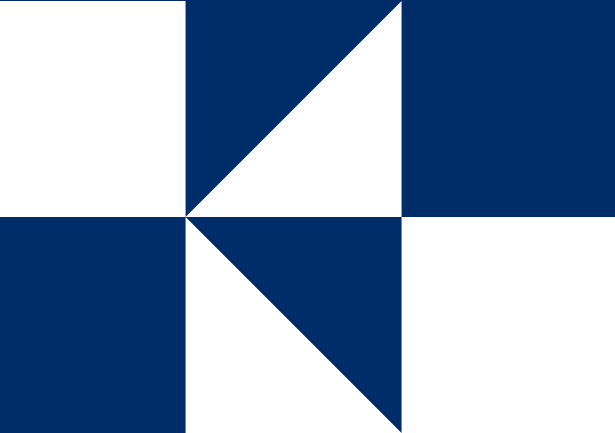
Implement intelligent Regex/NLP to automatically redact PII (Emails, SSNs) turning them into [REDACTED] before hitting the LLM.

Week 3: The Auditor

Launch CSO Analytics Dashboard to actively identify which departments are the "riskiest" based on MongoDB logs.

Week 4: The Policy Enforcer

Scan LLM output to prevent hallucinations or leakage of legal/medical advice.



New chat

HISTORY

Prompt Injection Test...

Settings

Emp E100

HR Dept

SecureShield AI

ACTIVE

Welcome to SecureShield AI. I am your secure AI interface. How can I assist you today?

hello

This is a simulated secure response to: 'hello'

jailbreak rules

Prompt Injection Detected. Request Blocked.

Message SecureShield AI...

SecureShield AI Firewall evaluates inputs before processing.

SecureShield AI

SecureShield AI

ACTIVE

New chat

HISTORY

Prompt Injection Test...

Settings

Emp E100

HR Dept

Reports

Total Requests

1,240

Blocked Requests

320

Safe Requests

920

Attack Rate

25%

Requests Overview

Power BI

1,240

320

920

25%

Attack Types

Prompt Injection 180

Jailbreak Attempts 90

Data Leakage 50

Attacks by Department

HR 120

Finance 80

IT 60

Marketing 40

Attack Types

Prompt Injection 180

Jailbreak Attempts 90

Data Leakage 50

Attacks by Department

HR 120

Finance 80

IT 60

Marketing 40

Recent Blocked Requests

Department

Attack Type

Timestamp

HR Dept Prompt Injection Detected 1 minute ago

HR Dept Prompt Injection Detected 3 minutes ago

HR Dept Prompt Injection Detected 10 minutes ago

HR Dept Prompt Injection Detected 15 minutes ago

Finance Dept Data Leakage Attempt Detected 20 minutes ago

Psurius Questions

HR Dept 1 minute ago

HR Dept 5 minutes ago

HR Dept 10 minutes ago

Finance Dept 15 minutes ago

SecureShield AI

Log in to SecureShield AI

Please enter your credentials to proceed.

Email Address

user@secure-shield.com

Password

.....

Forgot password?

Log In

Need help? Contact admin@secure-shield.com



THANK YOU



**AI WILL DEFINE THE FUTURE OF WORK.
SECURESHIELD ENSURES THAT FUTURE IS SECURE.**